

Titanic Data Analysis – 5-Minute Report

Project objective: explore the Titanic passenger data and identify descriptive survival patterns.

Dataset

The standard Titanic training dataset contains 891 passenger records and includes survival status, passenger class, sex, age, family-related fields, fare, cabin and embarkation port.

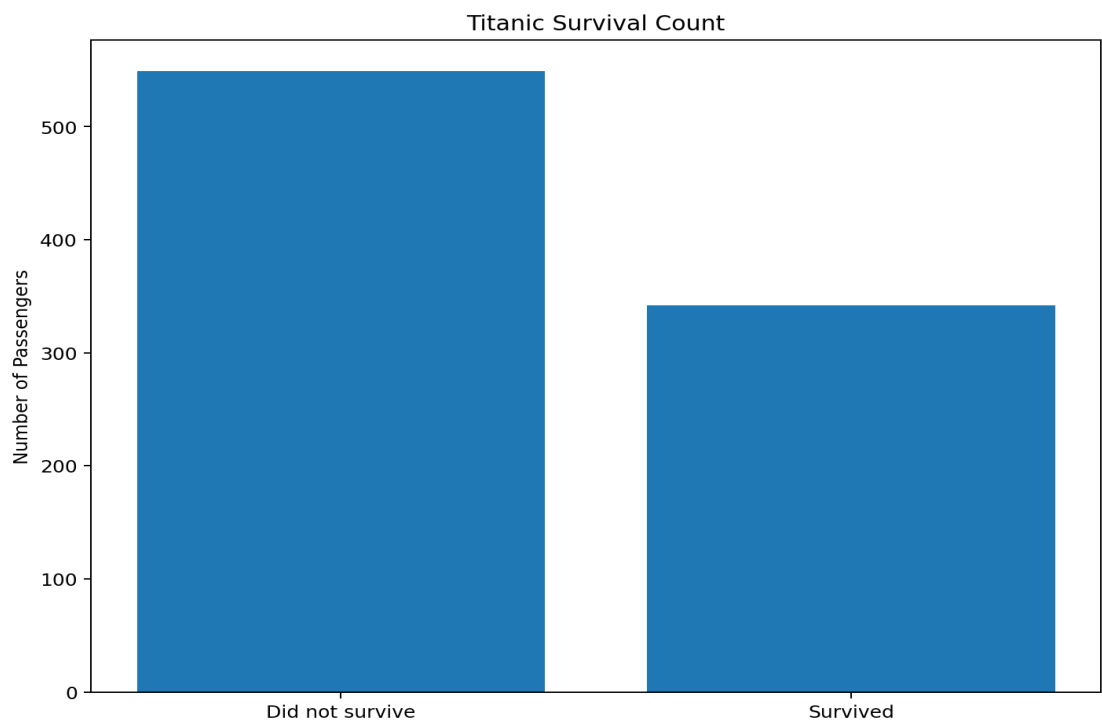
Key findings

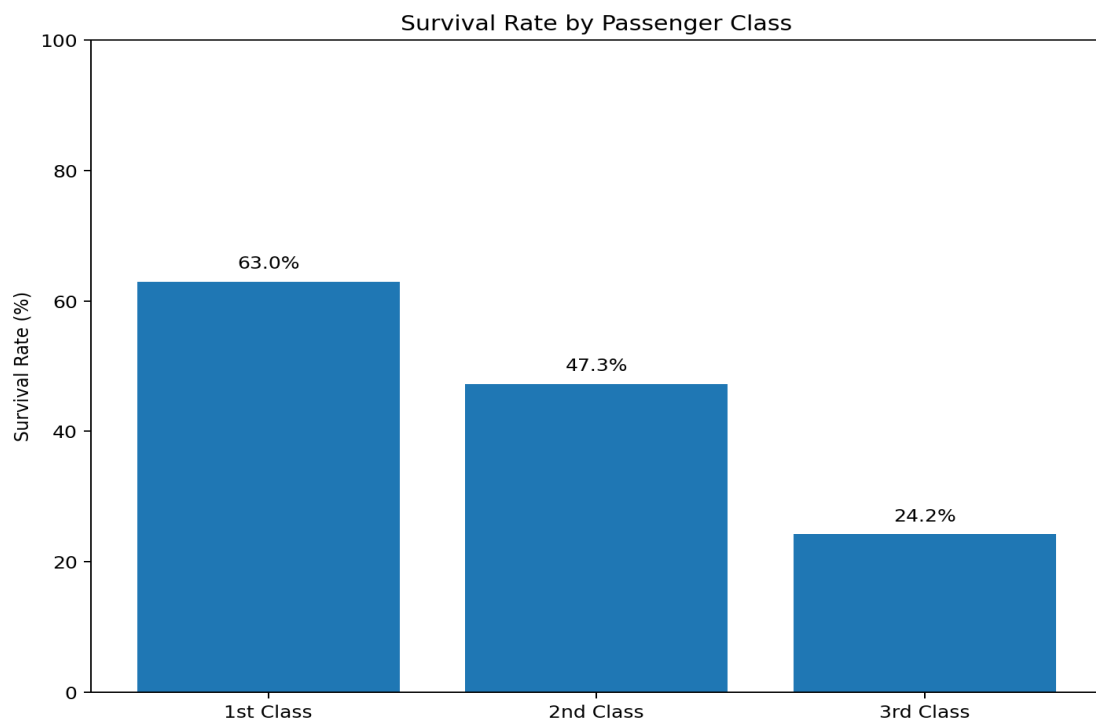
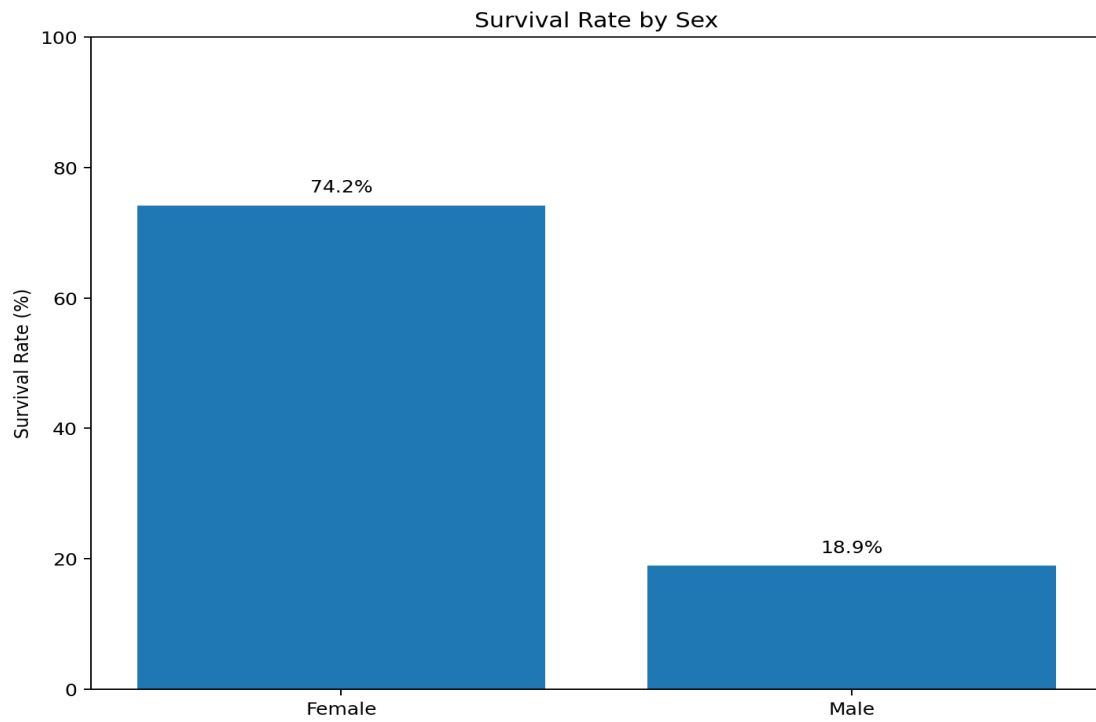
Metric	Result
Passengers	891
Survived	342 (38.4%)
Did not survive	549 (61.6%)
Female survival rate	74.2%
Male survival rate	18.9%
1st-class survival rate	63.0%
2nd-class survival rate	47.3%
3rd-class survival rate	24.2%

Interpretation

The descriptive results show substantial differences in observed survival rates across sex and passenger class. Female passengers had a much higher observed survival rate than male passengers in this dataset. Survival also varied by passenger class, with higher observed rates in first class than in second or third class. These are associations in the historical dataset and do not by themselves establish causation.

Visualizations





Conclusion

The project demonstrates how Pandas can be used for data cleaning and grouping, while Matplotlib and Seaborn can communicate survival patterns visually. The analysis can be extended with age groups, family size, fare bands and a machine-learning survival model.